



Climate Prediction Center

[home](#)[Site Map](#)[News](#)[Organization](#)[Search](#)[Go](#)[HOME](#) > [Climate & Weather Linkage](#) > [El Niño / Southern Oscillation \(ENSO\)](#)[Search](#)[Go](#)[Climate Outlooks](#)[Climate & Weather Link](#)[El Niño/La Niña](#)[MJO](#)[Teleconnections](#)[AO](#)[NAO](#)[PNA](#)[AAO](#)[Blocking](#)[Storm Tracks](#)[Climate Glossary](#)[Outreach](#)[About Us](#)[Our Mission](#)[Who We Are](#)[Contact Us](#)[Web Team](#)

El Niño - Southern Oscillation (ENSO)

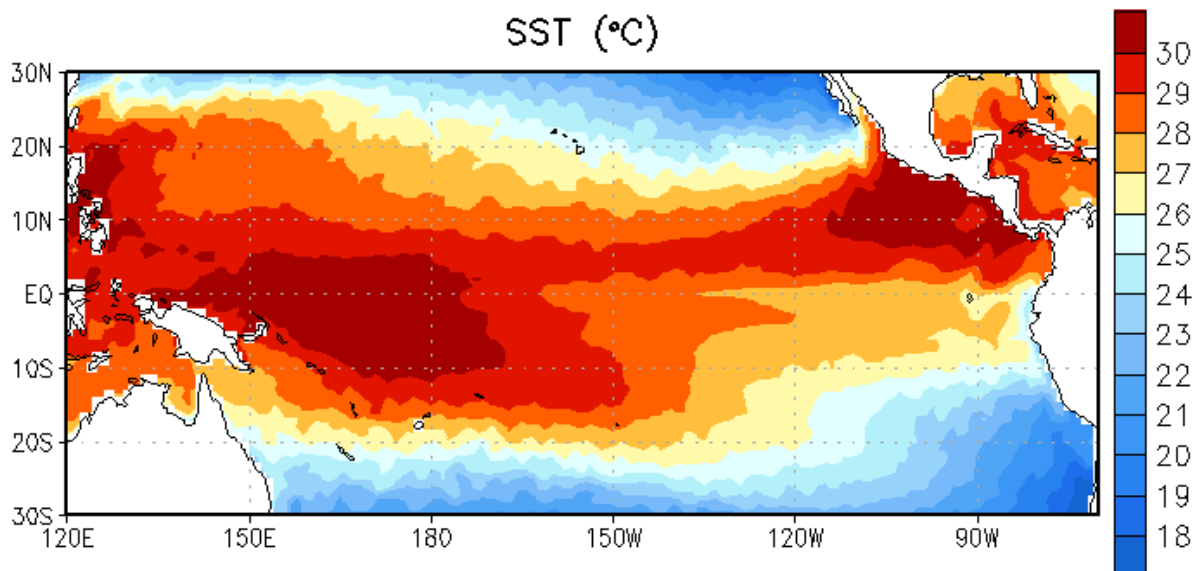
- [Current Conditions](#)
- [Historical Information](#)
- [Outlooks](#)
- [Expert Discussions/Assessments](#)
- [Diagnostic and Attribution Tools](#)
- [Educational Materials](#)
- [References](#)

Current Conditions

Note: Move cursor over product name to display the graphic.

Animations		
Sea Surface Temperatures (SST)	SST Anomalies - Tropical Pacific	T-Depth Anomalies - Equatorial Pacific
Tropical Outgoing Longwave Radiation (OLR) & Winds (Last 30 Days)		
OLR Anomalies	850hPa Wind Anomalies	200hPa Wind Anomalies
Sea Surface Temperatures (SST)		
Weekly Anomalies	Niño Regions Anomalies	Anomalies (5°N-5°S)

Week centered on 27 MAY 2026
SST (°C)



- [SST Animation](#) - Tropical Pacific

- [SST Animation](#) - Global
- [T-Depth Animation](#) - Equatorial Pacific

Outgoing Longwave Radiation (OLR)

[Pentad mean and anomalous OLR](#)

[Time-longitude section of anomalous OLR](#)

[Last 30 day average of anomalous OLR](#)

850-hPa Zonal Wind

[Time-longitude Section \(CORE data\)](#)

[Time-longitude Section \(Anomalous\)](#)

Sea Surface Temperature (SST)

[Weekly SST Anomalies](#)

[Time-longitude section of SST anomalies \(5°N-5°S\)](#)

[Time series of weekly SST anomalies for the 4 Niño regions](#)

[Time series of weekly SST and Climatology for the 4 Niño regions](#)

[Most recent Monthly SST Anomalies](#)

[Time series of monthly SST anomalies for the 4 Niño regions](#)

Subsurface Tropical Pacific Ocean Analyses

(Click [NCEP's Global Ocean Data Assimilation System](#) for more Information)

[Mean and anomalous equatorial temperatures](#)

[Equatorial depth of the 20°C isotherm](#)

[Anomalous equatorial depth of the 20°C isotherm](#)

[Equatorial 55 meter depth temperature anomalies](#)

[Equatorial 105 meter depth temperature anomalies](#)

[Equatorial 155 meter depth temperature anomalies](#)

[Sea level anomalies](#)

[Equatorial Upper-Ocean Temperature Anomalies](#)

[Most recent pentad of anomalous equatorial temperatures](#)

[\[Back to the Top\]](#)

■ Historical

- [Historical El Nino/ La Nina episodes \(1950–present\)](#)
- [Equatorial Upper-Ocean Heat Content Anomalies\(1979–present\)](#)

■ Outlooks

- [Official NOAA CPC ENSO Outlook](#)
- [Official NOAA CPC ENSO Probabilities](#)
- [Official NOAA CPC ENSO Strength Probabilities](#)
- [Seasonal CFSv2 SST Outlook](#)

[\[Back to the Top\]](#)

▪ Expert Discussions/Assessments

- [Weekly ENSO Evolution, Status, and Prediction Presentation \(PDF\)](#) 
- [Weekly ENSO Evolution, Status, and Prediction Presentation \(powerpoint\)](#) 
- [Weekly Global Tropics Benefits/ Hazards Assessment](#)
- [Monthly El Niño/La Niña Diagnostics Discussion](#)
- Monthly Climate Diagnostics Bulletin: [Current](#) [Archive](#)

[\[Back to the Top\]](#)

▪ Diagnostic and Attribution Tools

- [U.S. Temperature, Precipitation, and Snowfall Impacts based on historical ENSO episodes \(seasonal\)](#)
- [Global Temperature and Precipitation Linear Regressions and Correlations](#) 

[\[Back to the Top\]](#)

▪ Educational Material

Frequently asked questions area available to help the public better understand the climate system and how climate patterns in far off places affect our weather patterns.

• Fact Sheets

- [El Niño/La Niña](#) - Frequently Asked Questions on El Niño/La Niña.
- [El Niño/La Niña Cycle \(Tutorial\)](#) - El Niño/La Niña Tutorial.
- [El Niño and Climate Impacts](#) - Technical discussion of El Niño's oceanic and atmospheric conditions and their global climate impacts.
- [La Niña and Climate Impacts](#) - Technical discussion of La Niña's oceanic and atmospheric conditions and their global climate impacts.
- [SST Niño Regions](#) - Graphical depiction of the regions (i.e. "NINO Boxes") most commonly used in the diagnosis and forecast of El Niño.
- [Other El Niño Links](#) - Links to the most informative El Niño/La Niña links on the web.

- **Monographs** - The Office of Global Programs together with the University Consortium on Atmospheric Research in consultation with CPC and others produces a series of monographs written for the layperson called Reports to the Nation on Our Changing Planet. These monographs are used by many science teachers in their earth sciences classes.

- [El Niño and Climate Prediction](#)
- [Our Changing Climate \(text format\)](#)
- [Our Ozone Shield](#)

[\[Back to the Top\]](#)

▪ References

Barnston, A.G., C. F. Ropelewski, 1992: Prediction of ENSO episodes using canonical correlation analysis. J. Climate, 5, 1316-1345.

Barnston, A.G., M. Chelliah and S.B. Goldenberg, 1997: Documentation of a highly ENSO-related SST region in the equatorial Pacific. Atmosphere-Ocean, 35, 367-383.

Barnston, A. G., M. H. Glantz, and Y. He, 1999: Predictive skill of statistical and dynamical climate models in SST forecasts during the 1997-98 El Nino episode and the 1998 LNA Nina onset. *Bull. Am. Met. Soc.*, 80, 217-243.

Halpert, M. S. and C. F. Ropelewski, 1992: Surface temperature patterns associated with the Southern Oscillation. *J. Climate*, 5, 577-593.

Higgins, R.W., Y. Zhou and H.-K. Kim, 2001: Relationships between El Nino-Southern Oscillation and the Arctic Oscillation: A Climate-Weather Link. NCEP/Climate Prediction Center ATLAS 8.

Higgins, R. W., V. E. Kousky, H.-K. Kim, W. Shi, and D. Unger, 2002: High frequency and trend adjusted composites of United States temperature and precipitation by ENSO phase, NCEP/Climate Prediction Center ATLAS No. 10, 22 pp.

Hoerling, M. P., and A. Kumar. 1997. Why do North American climate anomalies differ from one El Nino event to another? *Geophysical Research Letters*, 24 (1 May): 1059-1062.

Hoerling, M. P., and A. Kumar, 1997: Origins of extreme climate states during the 1982-83 ENSO winter. *J. Climate*, 10, 2859-2870.

Hoerling, M. P., and A. Kumar, 2000: Understanding and predicting extratropical teleconnections related to ENSO. In *El Niño and the Southern Oscillation: Multi-scale Variations and Global and Regional Impacts*, H.F. Diaz and V. Markgraf (Eds.), Cambridge University Press, 57-88.

Hoerling, M. P., A. Kumar, and T. Xu, 2001: Robustness of the nonlinear climate response to ENSO's extreme phases. *J. Climate*, 14, 1277-1293.

Kousky, V.E., M.T. Kayano, and I.F.A. Cavalcanti, 1984: The Southern Oscillation: Oceanic-atmospheric circulation changes and related rainfall anomalies. *Tellus*, 36A, 490-504.

Kousky, V. E. 1997. Warm (El Nino) episode conditions return to the tropical Pacific. *Mariners Weather Log* 41, no.1 (Spring): 4-7.

Kousky, V. E. and R. W. Higgins, 2004: An Alert Classifications System for Monitoring and Assessment of the ENSO Cycle, *Wea and Forecasting*, **22**, 353-371

Kumar, A., and M. P. Hoerling, 1997: Interpretation and implications of observed inter-El Niño variability. *J. Climate*, 10, 83-91.

Kumar, A., and M. P. Hoerling, 1998: Annual cycle of Pacific/North American seasonal predictability associated with different phases of ENSO. *J. Climate*, 11, 3295-3308.

Kumar, A., A. Barnston, P. Peng, M. P. Hoerling, and L. Goddard, 2000: Changes in the spread of the seasonal mean atmospheric states associated with ENSO. *J. Climate*, 13, 3139-3151.

Kumar, A., and M. P. Hoerling, 2003: The nature and causes for the delayed atmospheric response to El Niño. *J. Climate*, 16, 1391-1403.

Leetma, Ants. 1989. The interplay of El Nino and La Nina. *Oceanus*, 32 (Summer) : 30-3434

Leetmaa, A. 1999: The first El Niño observed and forecasted from start to finish. *Bull. Am. Met. Soc.*, 80, 111-112.

Rasmusson, E. M. and T. H. Carpenter, 1982: Variations in tropical sea surface temperature and surface wind fields associated with the Southern Oscillation / El. Nino. *Mon. Wea. Rev.*, 110, 354-384.

Rasmusson, E. M., T. H. Carpenter, 1983: The relationship between eastern equatorial Pacific sea surface temperatures and rainfall over India and Sri Lanka. *Mon. Wea. Rev.*, 111, 517-528.

Rasmusson, E. M., and J. M. Wallace, 1983: Meteorological aspects of El Nino / Southern Oscillation. *Science*, 222, 1195-1202.

Rasmusson, E. M. and K. Mo, 1993: Linkages between 200-mb tropical and extratropical circulation anomalies during the 1986-1989 ENSO cycle. *J. Climate*, 6, 595-616.

Ropelewski, C. F., and M. S. Halpert, 1986: North American precipitation and temperature patterns associated with the El Nino/ Southern Oscillation (ENSO). Mon. Wea. Rev., 114, 2352-2362.

Ropelewski, C. F., and M. S. Halpert ,1987: Global and regional scale precipitation patterns associated with the El Nino/ Southern Oscillation. Mon. Wea. Rev., 115, 1606-1626.

Ropelewski, C.F. and P.D. Jones, 1987: An extension of the Tahiti-Darwin Southern Oscillation Index. Mon. Wea. Rev, 115, 2161-2165.

Ropelewski, C. F., and M. S. Halpert, 1989: Precipitation patterns associated with the high index phase of the Southern Oscillation. J. Climate, 2, 268-284.

Ropelewski, C. F., and M. S. Halpert, 1996: Quantifying Southern Oscillation-precipitation relationships. J. Climate, 9, 1043-1059.

Xue, Y. and A. Leetmaa, 2000: Forecast of tropical pacific SST and sea level using a Markov model. accepted by Geophys. Res. Lett.

Xue, Y., A. Leetmaa, and M. Ji, 2000: ENSO prediction with Markov models: the impact of sea level. J. Climate, 13, 849-871.

[\[Back to the Top\]](#)